



Hiring Assignment

Hiring: Full-Time Full Stack Developer (MERN + React Native)
Confidential — For Applicants Only

Objective

Build a simplified event ticket booking flow focused on seat reservation and booking confirmation.

Backend (Nodejs + Express + MongoDB)

Data Model (MongoDB)

- Event with basic fields (name, date/time, venue, total seats)
- Seat with eventId, seatNumber, and status (available, reserved, booked)
- Reservation with userId, eventId, seatNumbers, expiresAt

API Endpoints (Express.js)

- GET /api/events - Retrieve a list of all available events.
- GET /api/events/:id - Retrieve details for a single event.
- POST /api/reserve - Reserve available seats for a specific event for 10 minutes.
- POST /api/bookings - Marks the seats as booked and removes the reservation

Constraints

- Prevent double booking using atomic operations or transactions.
- Expired reservations should not allow booking.

Frontend (React.js)

- The frontend must provide a clean, responsive user interface to interact with the backend API.

Flow:

1. List all events
2. On selecting an event, show a seat grid with color coding for statuses and allow selecting multiple seats.
3. Click **Reserve** → calls /api/reserve and shows a countdown timer for the reservation
4. **Confirm Booking** → calls /api/bookings and shows success or failure
5. Display errors when a seat becomes unavailable between selection and reservation/booking

Key Technical Expectations:

- Effective state management (e.g., using React Hooks like useState and useEffect).
- Demonstration of component-based architecture.
- Handling of asynchronous API calls (e.g., using fetch or axios).
- Error Handling and Data Validation
- Basic User Authentication
- Include basic README with: how to run backend and frontend, any assumptions, and a short note on design decisions (e.g., how you avoid double booking)